



JKBOSE 2026

CLASS 11TH

STRAIGHT LINES

MATHS

LECTURE-02

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TOPICS TO BE COVERED

1

Slope of a Line

2

Angle between two Lines

3

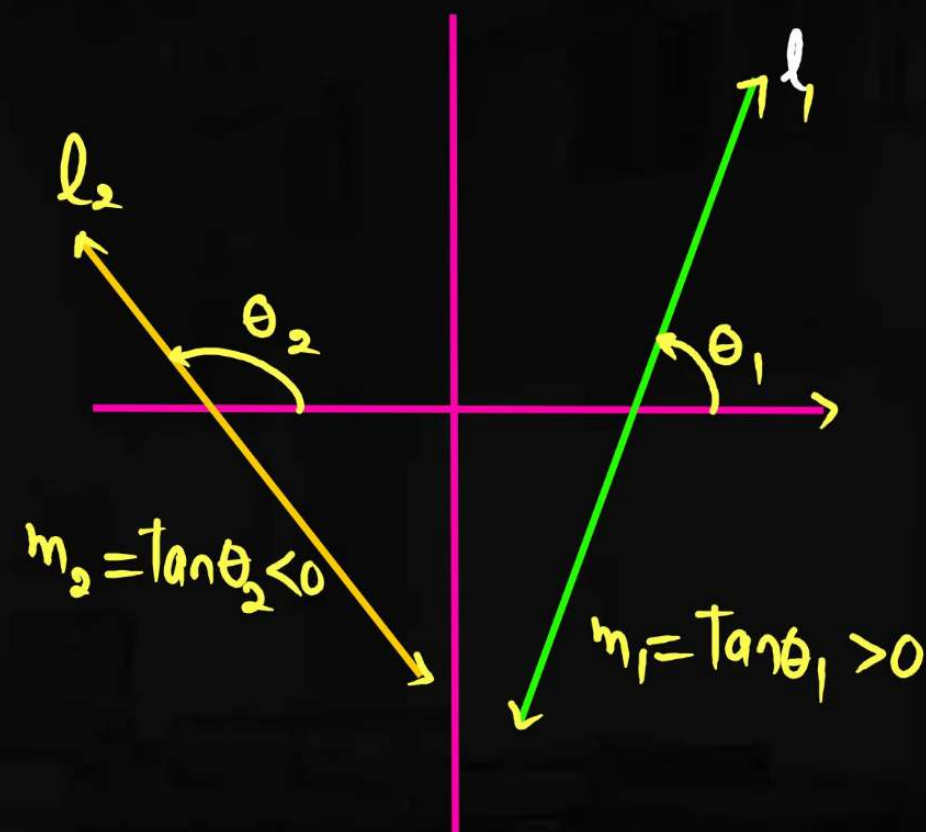
Exercise Discussion 9.1

4

Related Problems

Slope of a Line

- * The angle θ (Measured anticlockwise) which the line l makes with the positive direction of x -axis is known as Inclination angle / angle of Inclination
- * Obviously $0 \leq \theta < \pi$
- * If θ is the angle of inclination, then $\tan \theta$ is known as the slope of the line (Gradient of a line)
- * Generally denoted by m . So $m = \tan \theta$



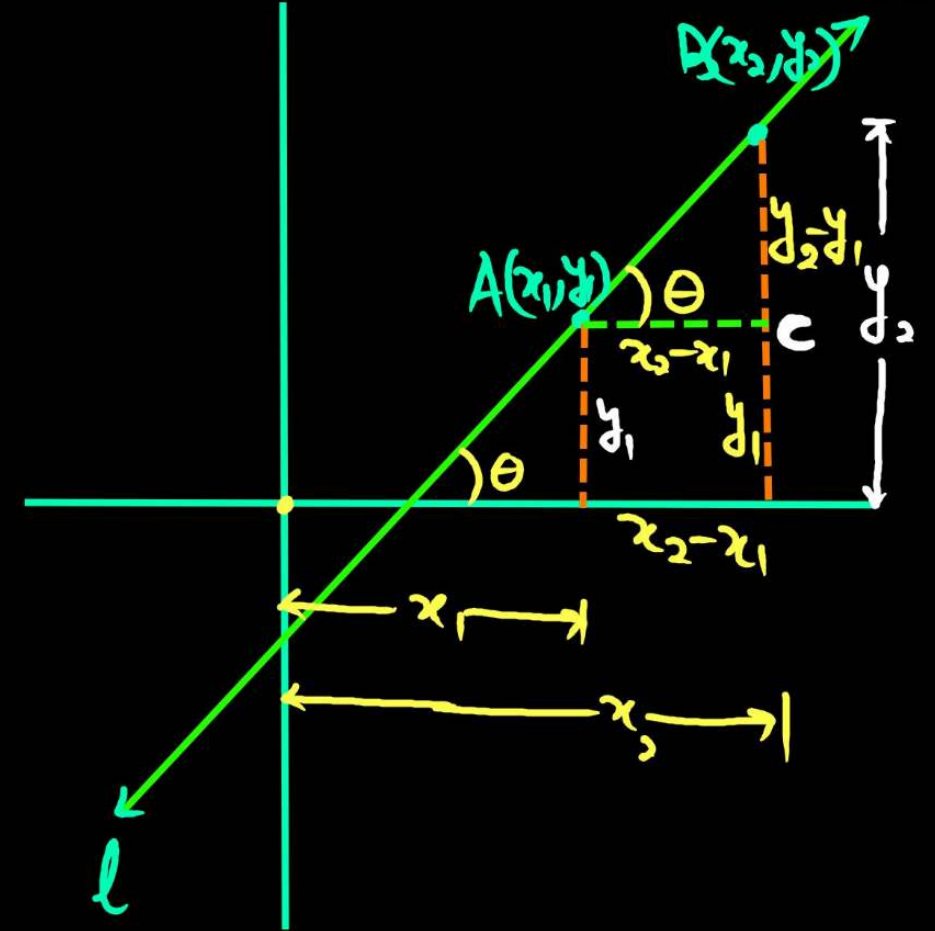
Slope of a line passing through two points:

Suppose that the line l passes through the points $A(x_1, y_1)$ & $B(x_2, y_2)$ with θ be its inclination angle.

We know that, $m = \tan \theta$

$$\Rightarrow m = \frac{BC}{AC}$$

$$\Rightarrow m = \frac{y_2 - y_1}{x_2 - x_1}$$



EXAMPLE

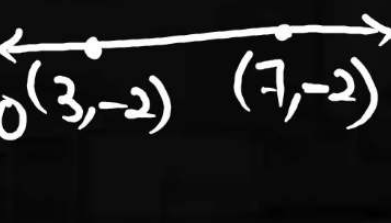
Find the slope of the lines:

- (a) Passing through the points $(3, -2)$ and $(-1, 4)$,
- (b) Passing through the points $(3, -2)$ and $(7, -2)$,
- (c) Passing through the points $(3, -2)$ and $(3, 4)$, H/W
- (d) Making inclination of 60° with the positive direction of x -axis.

(Sol) (a) 

$$\text{Required slope} = \frac{4+2}{-1-3} = \frac{6}{-4}$$

$$\Rightarrow \text{Required slope} = -\frac{3}{2}$$

(b) Required slope = $-\frac{2+2}{7-3} = 0$ 

(d) Required slope = $\tan 60^\circ = \sqrt{3}$.

EXAMPLE

A ray of light coming from the point $(1, 2)$ is reflected at a point A on the x -axis and then passes through the point $(5, 3)$. Find the coordinates of the point A .

(Sol) Here $m_{AQ} = \frac{3-0}{5-a} = \frac{3}{5-a}$

$$\Rightarrow \tan \theta = \frac{3}{5-a} \quad \text{--- (1)}$$

Also $m_{AP} = \frac{2-0}{1-a} = \frac{2}{1-a}$

$$\Rightarrow \tan(\pi - \theta) = \frac{2}{1-a}$$

$$\Rightarrow -\tan \theta = \frac{2}{1-a} \quad \text{--- (2)}$$

Using (1) in (2), we get;

$$\frac{-3}{5-a} = \frac{2}{1-a}$$

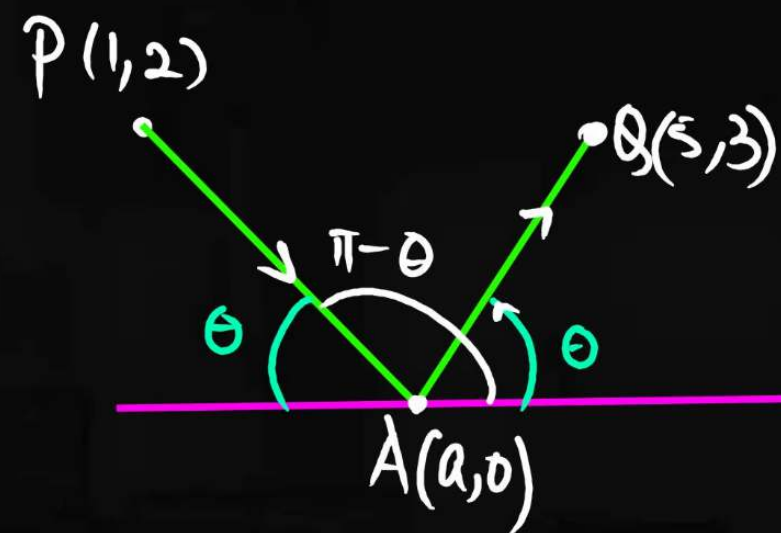
$$\Rightarrow -3 + 3a = 10 - 2a$$

$$\Rightarrow 5a = 13$$

$$\Rightarrow a = \frac{13}{5}$$

So, the required point is $A\left(\frac{13}{5}, 0\right)$

Hence the result



EXAMPLE

If the line joining two points $A(2, 0)$ and $B(3, 1)$ is rotated about A in anticlockwise direction through an angle of 15° . Find the slope of the line in new position.

(Sol) Here $m_{AB} = \frac{1-0}{3-2} = 1$.

$\Rightarrow \tan \theta = 1$

$\Rightarrow \theta = 45^\circ$

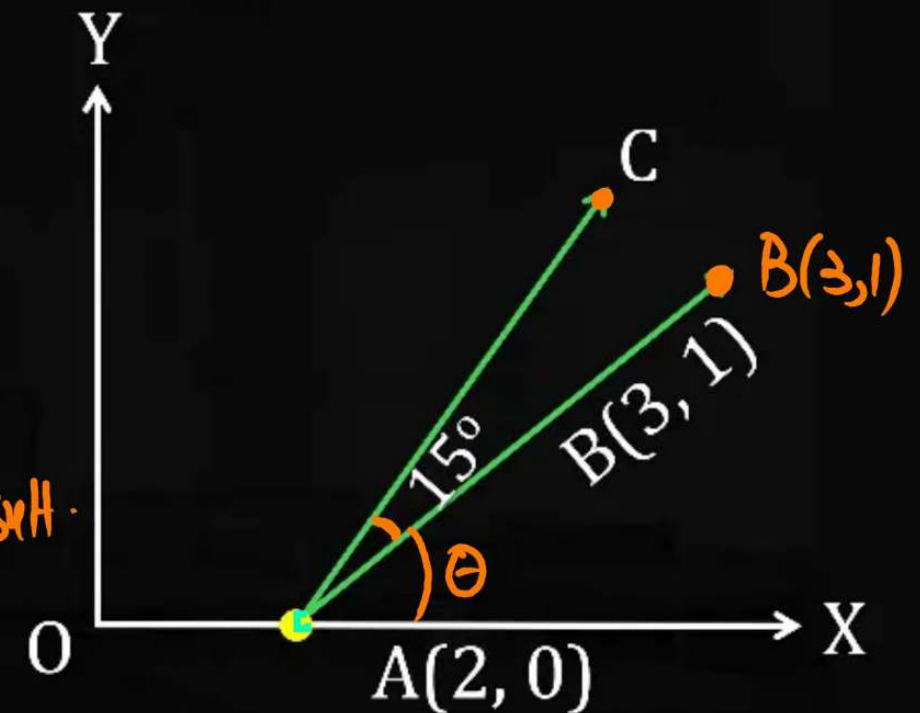
So, slope of line in the new position is given by;

$m_{AC} = \tan(\theta + 15^\circ) = \tan(45^\circ + 15^\circ)$

$\Rightarrow m_{AC} = \tan 60^\circ$

$\Rightarrow m_{AC} = \sqrt{3}$

Hence the result.



Angle Between Two Lines

Suppose that θ (Acute angle) is the angle b/w the lines l_1 & l_2 . Then;

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

Case 1st: If l_1 is parallel to l_2 , then $\theta = 0$

$$\Rightarrow \tan \theta = 0$$

$$\Rightarrow \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| = 0$$

$$\Rightarrow \frac{m_2 - m_1}{1 + m_1 m_2} = 0$$

$$\Rightarrow m_2 - m_1 = 0$$

$$\Rightarrow m_2 = m_1$$

So the slopes of parallel lines are equal.

Case 2nd:

If l_1 is perpendicular to l_2 , then

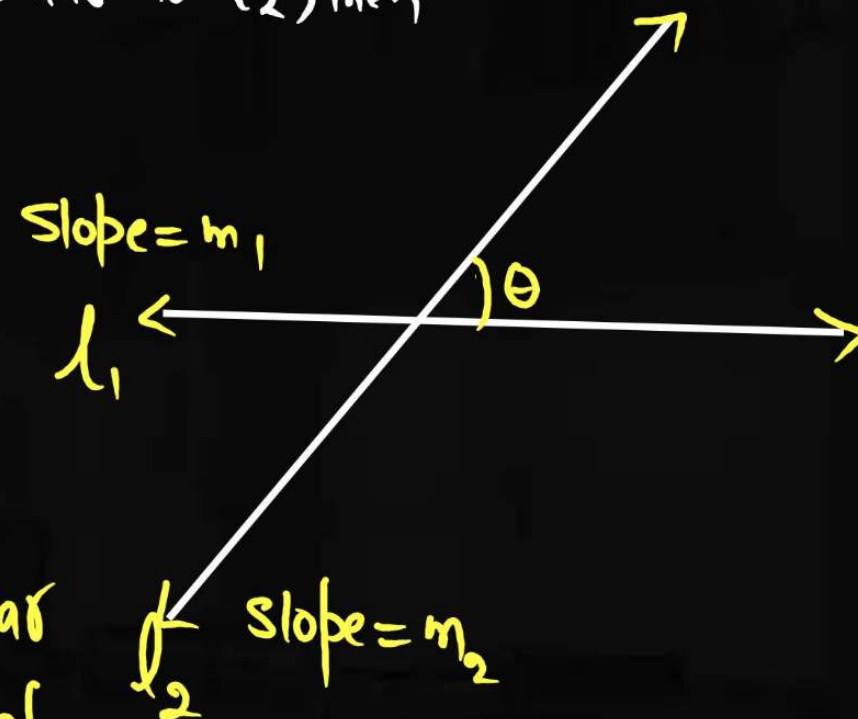
$$\theta = 90^\circ$$

$\Rightarrow \tan \theta$ is not defined, so

$$1 + m_1 m_2 = 0$$

$$\Rightarrow m_1 m_2 = -1$$

So, for perpendicular lines, the product of slopes is -1 .



EXAMPLE

If the angle between two lines is $\frac{\pi}{4}$ and slope of one of the lines is $\frac{1}{2}$, find the slope of the other line.

(Sol) we know that

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| = \left| \frac{m - \frac{1}{2}}{1 + \frac{m}{2}} \right|$$

$$\Rightarrow \tan\left(\frac{\pi}{4}\right) = \left| \frac{2m-1}{2+m} \right| = \left| \frac{2m-1}{m+2} \right|$$

$$\Rightarrow \left| \frac{2m-1}{m+2} \right| = 1.$$

$$\Rightarrow \frac{2m-1}{m+2} = 1, -1$$

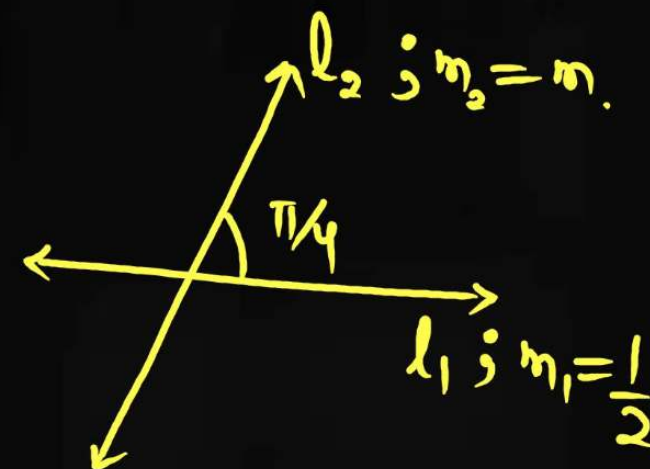
$$\Rightarrow \frac{2m-1}{m+2} = 1, \frac{2m-1}{m+2} = -1$$

$$\Rightarrow 2m-1 = m+2; 2m-1 = -m-2$$

$$\Rightarrow m = 3, 3m = -1$$

$$\Rightarrow m = 3, -\frac{1}{3}.$$

So, the required slopes are $3, -\frac{1}{3}$. Hence the result.



EXAMPLE

Line through the points $(-2, 6)$ and $(4, 8)$ is perpendicular to the line through the points $(8, 12)$ and $(x, 24)$. Find the value of x .

(Sol) Since we know that for perpendicular lines, the product of slopes is -1

$$\text{So, } m_1 \times m_2 = -1$$

$$\Rightarrow \left(\frac{8-6}{4+2} \right) \times \left(\frac{12-24}{8-x} \right) = -1$$

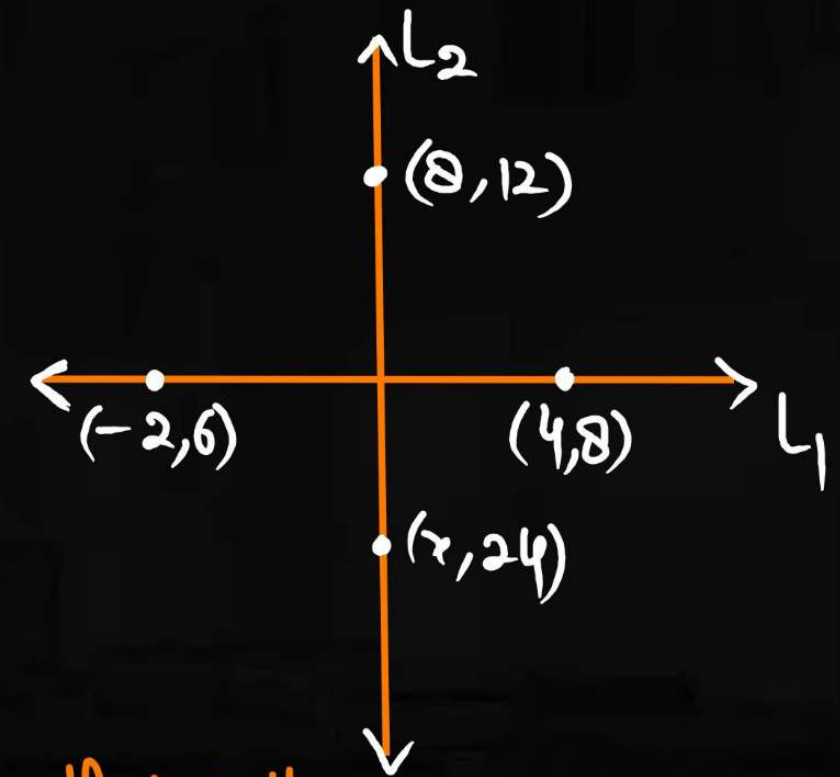
$$\Rightarrow \frac{2}{6} \times \frac{-12}{(8-x)} = -1$$

$$\Rightarrow \frac{4}{8-x} = 1$$

$$\Rightarrow 4 = 8 - x$$

$$\Rightarrow x = 8 - 4$$

$$\Rightarrow \boxed{x = 4}$$



Hence the result.

QUESTION

If the line segments AB and CD are parallel, then the value of k is:

1

2 ✗

2

4 ✓

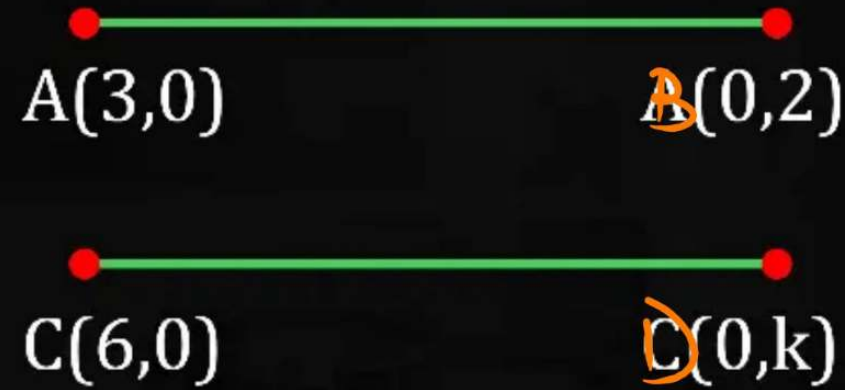
3

8 ✗

4

3 ✗

$$\begin{aligned}
 m_{AB} &= m_{CD} \\
 \Rightarrow \frac{2-0}{0-3} &= \frac{k-0}{0-6} \\
 \Rightarrow \frac{2}{-3} &= \frac{k}{-6} \\
 \Rightarrow k &= \frac{2}{-3} \times -6 \\
 \Rightarrow k &= 4
 \end{aligned}$$



QUESTION

Two lines have slopes $m_1 = 3$ and $m_2 = \frac{1}{2}$, then the tangent of the acute angle between them is :

1 $\frac{5}{2}$ ✗

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

2 1 ✓ $\Rightarrow \tan \theta = \left| \frac{\frac{1}{2} - 3}{1 + \frac{3}{2}} \right| = \left| \frac{\frac{1-6}{2}}{\frac{2+3}{2}} \right| = \left| \frac{-5}{5} \right| = 1$

3 $\frac{1}{2}$ ✗

$$\tan \theta = 1$$

4 0 ✗

QUESTION

If the angle between two lines is 45° and the slope of one line is 2, then the sum of the square of the possible values of the slope of the second line is :

1

$$\frac{82}{3} \times$$

(Sol) we know that,
 $\tan 45^\circ = \left| \frac{m-2}{1+2m} \right|$

2

$$\frac{82}{9} \checkmark$$

$$\Rightarrow \left| \frac{m-2}{1+2m} \right| = 1$$

3

$$9 \times$$

$$\Rightarrow \frac{m-2}{1+2m} = 1, -1$$

4

$$\frac{1}{9} \times$$

$$\Rightarrow \frac{m-2}{1+2m} = 1, \frac{m-2}{1+2m} = -1$$

$$\Rightarrow m-2 = 1+2m; m-2 = -1-2m$$

$$\Rightarrow m-2m = 1+2; m+2m = -1+2$$

$$\Rightarrow m = -3, \frac{1}{3}$$

Therefore Required value $= (-3)^2 + \left(\frac{1}{3}\right)^2$
 $= 9 + \frac{1}{9}$
 $= \frac{82}{9}$

QUESTION

If the slope of the line **L_1 is m** and **line L_2 is vertical**, the angle **α** that L_1 makes with L_2 is given by :

1 $\tan \alpha = |m|$

(H/W)

2 $\sin \alpha = m$

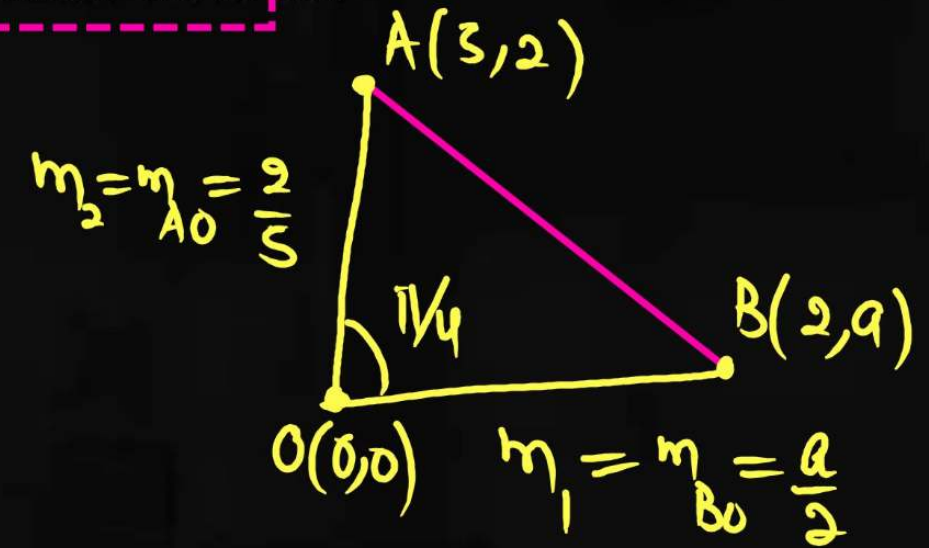
3 $\alpha = 90^\circ$

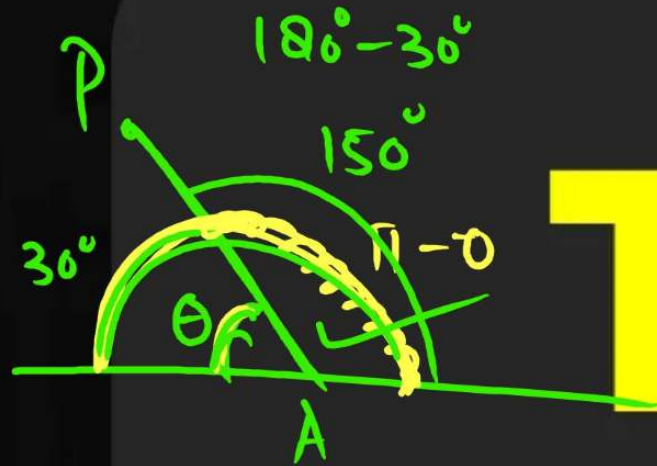
4 $\cot \alpha = |m|$

QUESTION (JEE MAINS 2024):

If the line segment joining the points $(5, 2)$ and $(2, a)$ subtends an angle $\frac{\pi}{4}$ at the origin, then the absolute value of the product of all possible values of a is:

- 1 6
- 2 8
- 3 2
- 4 4





Thank
You